# Highlights

## Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample subsurface environments.
- Event-specific ERA5–HYCOM fields cover 500 historical TCs.
- Parameter-sharing IPPO scales TC tracking across 4–16 SUVs.
- Current and future SUV configurations are compared at both stages.

# Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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## ARTICLE INFO

*Keywords*:
tropical cyclone observation
cooperative task
submersible unmanned vessel
multi-agent reinforcement learning
surface–subsurface observation

## ABSTRACT

Tropical cyclones are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Conventional platforms rarely combine rapid transit with surface-wind and subsurface-current sensing. Blue Whale-class submersible unmanned vessels can transit rapidly at the surface to TC cores, then dive to observe temperature, salinity, currents, and biogeochemical properties. We use simulation-based multi-agent reinforcement learning for autonomous navigation and sampling. The environments combine ERA5 10 m winds and HYCOM currents from 0 to 60 m with IBTrACS tracks for 500 western North Pacific TCs. A parameter-sharing independent proximal policy optimization (IPPO) policy maps shared surface-wind observations or local current histories to continuous control commands at 10 min intervals. The surface policy assumes idealized global sharing of observed wind information, whereas the underwater policy uses no inter-vehicle communication. We compare 12 m/s surface and 1.5 m/s submerged limits, close to current Blue Whale design speeds, with 20 m/s and 2.0 m/s as future upgrades. At the surface, the fleet follows the TC toward high-wind regions despite uncertain deployment, reducing final tracking error and improving observation benefit. Underwater, active control outperforms random control and passive drift in mean experienced current, trajectory extent, and spatial coverage. Compared with four traditional baselines, IPPO achieves better surface tracking distances. Underwater, potential field has the highest mean experienced current, while IPPO gives the largest nominal and current-weighted coverage. The simulation demonstrates wind-guided surface approach and current-guided fixed-depth sampling by coordinated SUV fleets.

## 1. Introduction

Tropical cyclones (TCs) threaten coastal communities, infrastructure, and offshore operations. Climate assessments indicate that some TC changes are detectable despite observational uncertainty [1] and project an increasing proportion of very intense TCs under continued warming [2], while global TC frequency may decline [3] and rapid-intensification events have recently increased in offshore regions [4]. Landfalling TCs over the western North Pacific and China have also become more destructive, with increasing nearshore intensification and exposure [5, 6, 7, 8]. Track forecasts have improved substantially, but intensity prediction remains more difficult: recent intensity guidance has reduced errors [9], yet official forecasts in the western North Pacific continue to carry substantial uncertainty [10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12].

Existing observing platforms each cover only part of this problem. Spaceborne microwave sensors provide broad but complementary and incomplete views of the near-core wind field. Active scatterometers infer surface wind from ocean backscatter, but lose sensitivity at high wind speeds and can be degraded by rain-induced attenuation and scattering [13]. Passive radiometers infer wind from ocean emissivity and can penetrate some cloud cover, but cloud liquid water, precipitation emission, and atmospheric attenuation contaminate the brightness-temperature signal [13, 14]. Neither sensor class resolves the subsurface ocean [15]. Aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [16, 17], whereas gliders remain at sea for long periods but move much more slowly than a translating TC [18]. AUVs can adapt their sampling to changing environmental and vehicle states [19]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [20, 21, 22, 23, 24], but their value is highly sensitive to sampling location and background structure. Targeted observation studies likewise show that forecast benefit depends on sensitive regions, the background state, and the sampling design [25, 26, 27]. These limitations leave near-core wind and upper-ocean states poorly constrained in many forecast initial conditions. Adaptive TC observation requires decisions about motion, sampling time, and coordination, not coverage alone.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [28, 29, 30, 31]. The operational concept assumes that the SUVs are deployed

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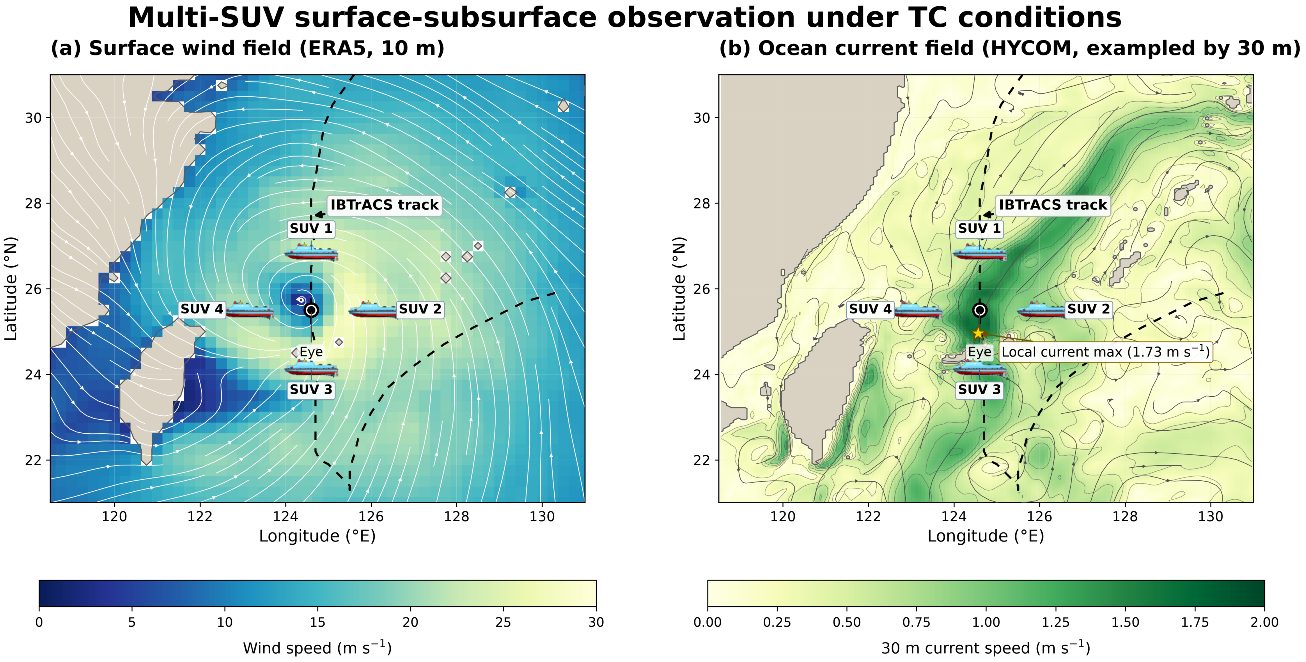
Multi-SUV surface-subsurface observation under TC conditions
(a) Surface wind field (ERA5, 10 m)
(b) Ocean current field (HYCOM, exampled by 30 m)
IBTrACS track
SUV 1
SUV 2
SUV 3
SUV 4
Eye
Local current max (1.73 m s$^{-1}$)
Latitude (°N)
Longitude (°E)
Wind speed (m s$^{-1}$)
30 m current speed (m s$^{-1}$)

**Figure 1:** Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

from a mother ship, such as a large research vessel or an equivalent offshore support ship, or from an offshore floating platform such as the Shanghai Jiao Tong University Far-Seas Floating Island (Yuanhai Fudongdao). Such platforms can provide rapid relocation, long-duration station keeping, and deployment of heavy marine equipment [32, 33, 34]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and an observational payload that includes an acoustic Doppler current profiler and a CTD sensor, with a public reference depth of approximately 60 m [28]. This study uses the current Blue Whale's surface–subsurface motion capabilities as the reference parameterization and evaluates a possible future-upgrade configuration as a comparison case. The vehicles are simplified as point masses for path planning. In the proposed mission, the surface fleet observes 10 m winds, follows the TC toward the high-wind region as closely as possible, and is evaluated by its final tracking error. At depth, four SUVs are assigned to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this surface-to-subsurface task switch; four vehicles are shown at the surface as an example, and the same parameter-sharing policy is later evaluated with 4, 8, 12, and 16 SUVs.

Traditional graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [35, 36, 37, 38, 39, 40]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [41, 42, 43, 44, 45, 46, 47, 48], and independent PPO is competitive in cooperative multi-agent tasks [49, 50, 51]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [52, 53, 54, 55, 56].

Three gaps remain. First, a unified atmosphere–ocean reinforcement-learning environment for coordinated surface–subsurface TC observation has not been established [57, 58]. Second, it remains unclear whether parameter-sharing MARL can track a moving TC under deployment uncertainty, sample the subsurface current field, and remain effective across different fleet sizes [25, 26, 27, 58]. Third, learned policies have rarely been compared with causal planners under identical local observations, motion constraints, and control interfaces. We formulate stochastic policy optimization under partial observability [59, 49, 50], evaluate 4-, 8-, 12-, and 16-SUV fleets, and provide this matched comparison.

The main contributions of this study are summarized as follows.

1. We construct a unified atmosphere–ocean RL environment by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS

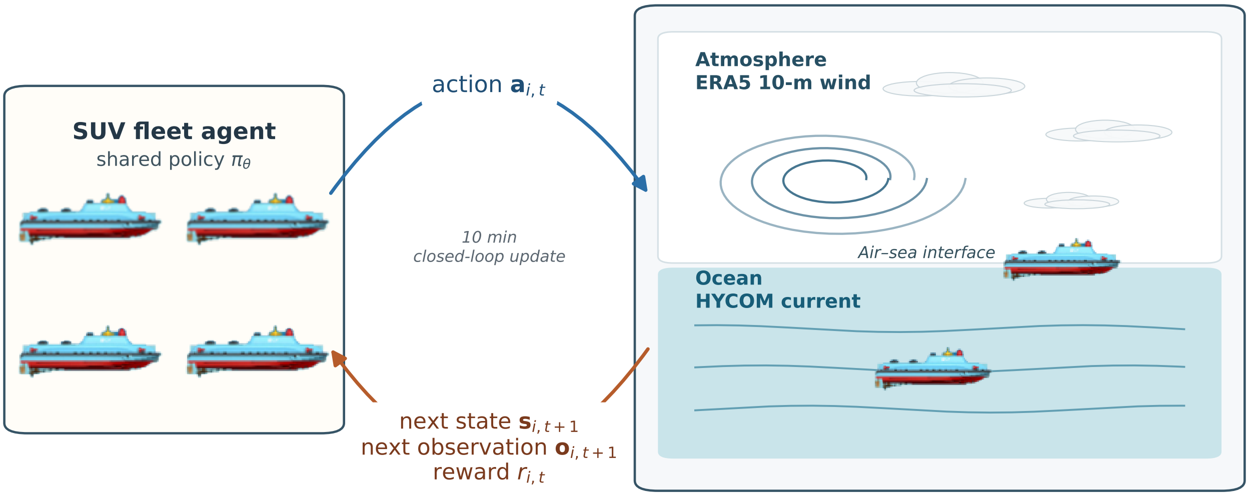
SUV fleet agent
shared policy $\pi_\theta$
action $\mathbf{a}_{i,t}$
10 min
closed-loop update
Atmosphere
ERA5 10-m wind
Air–sea interface
Ocean
HYCOM current
next state $\mathbf{s}_{i,t+1}$
next observation $\mathbf{o}_{i,t+1}$
reward $r_{i,t}$

**Figure 2:** Closed-loop interaction between the SUV fleet agent and the tropical-cyclone atmosphere–ocean environment. At each 10 min interval, the SUV fleet sends actions generated by its shared policy $\pi_\theta$ to the environment, which returns the next state, next observation, and reward to the fleet.

tracks. The 500-TC archive is divided at the TC level into 450 training TCs and 50 held-out evaluation TCs.

2. We evaluate a parameter-sharing IPPO controller with globally shared surface-wind observations and decentralized subsurface current histories for 4-, 8-, 12-, and 16-SUV fleets. Surface policies are tested at 100–800 km deployment offsets, and subsurface policies are tested at seven depths and two propulsion limits. We report final and time-averaged tracking distance, experienced current, nominal coverage, and current-weighted coverage.
3. We compare IPPO with four traditional planners, including artificial potential field, receding-horizon A*, lawnmower coverage, and information-greedy control, under matched local observations, action constraints, motion constraints, control intervals, and paired evaluation episodes. At the surface, IPPO achieves lower final and time-averaged tracking distance than the planners and generally uses shorter paths. Underwater, potential field achieves the highest mean experienced current, whereas IPPO achieves the largest nominal and current-weighted coverage.

## 2. Problem Formulation

### 2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, each SUV navigates toward the predicted TC center and high-wind region, with the final tracking error used to evaluate how closely the fleet follows the TC. In the operational concept, the fleet would descend after approaching a predefined high-wind observation window. The learning experiments evaluate this mission as two separate episodes: a surface approach episode and a subsurface sampling episode in which four SUVs seek the highest attainable current speed at each prescribed depth. The mission imposes four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. decentralized tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under action smoothing and coastline safeguards.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reach the high-wind region as closely as possible, quantified primarily by the final tracking error. At depth, the primary learning objective changes to finding and sampling higher-current water, while nominal and current-weighted spatial coverage characterize how broadly and how valuably the fleet samples the dynamically active region. The four SUVs must balance active propulsion against passive advection by the ambient current.

### 2.2. Problem Definition

The fleet contains $N$ SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \dots, N\}$. The position, commanded horizontal velocity, and operating mode of SUV $i$ at time $t$ are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. Horizontal positions and reported distances are expressed in km, whereas commanded and ocean-current velocities are expressed in m/s. The commanded speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V_{\max}^{\mathrm{s}}, & m_i(t) = \mathrm{s}, \\ V_{\max}^{\mathrm{u}}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. The mode $m_i(t)$ selects the applicable speed bound and motion law for that episode; continuous diving, ballast transitions, and resurfacing are not simulated within a single decision trajectory.

For the motion equations, let $\mathbf{p}_i$ denote the desired horizontal velocity command and $\mathbf{q}_i^{\mathrm{prop}}$ the effective propulsion velocity after the acceleration and heading dynamics. The km/min representation of the ocean current is denoted by $\mathbf{v}_{\mathrm{ocean}}$; the device-level speed bounds above remain reported in m/s. The surface uses the effective propulsion velocity directly as ground velocity, whereas the underwater stage adds current advection. The horizontal motion in km is

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{q}_i^{\mathrm{prop}}(t), & m_i(t) = \mathrm{s}, \\ \mathbf{q}_i^{\mathrm{prop}}(t) + \mathbf{v}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m. Time $t$ is measured in minutes, and $\mathbf{v}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \mathbf{x}_{i,k} + \Delta t \begin{cases} \mathbf{q}_{i,k}^{\mathrm{prop}}, & m_{i,k} = \mathrm{s}, \\ \mathbf{q}_{i,k}^{\mathrm{prop}} + \mathbf{v}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k), & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

Here $\Delta t = 10$ min. The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

The unit-disk action defines a desired speed and course. In the surface stage, the command specifies a desired horizontal ground velocity, which is tracked by the acceleration and heading dynamics without an added current term. In the underwater stage, the command specifies a desired propulsion velocity before current advection is applied:

$$\begin{aligned} \mathbf{p}_{i,k}^{\mathrm{des}} &= V_{\max}^{m_i} \bar{\mathbf{a}}_{i,k}, \qquad s_{i,k}^{\mathrm{des}} = \left\| \mathbf{p}_{i,k}^{\mathrm{des}} \right\|_2, \\ \psi_{i,k}^{\mathrm{des}} &= \mathrm{atan2}\left(\bar{a}_{i,k,y}, \bar{a}_{i,k,x}\right). \end{aligned} \tag{4}$$

Each SUV carries an effective propulsion speed $s_i$ and heading $\psi_i$, with $\mathbf{q}_i^{\mathrm{prop}} = s_i[\cos\psi_i, \sin\psi_i]^\top$. The 10 min control interval is divided into ten 60 s substeps. In substep $\ell$, speed and heading evolve according to

$$\begin{aligned} s_i^{\ell+1} &= \mathrm{clip}\big(s_i^{\ell} \\ &\quad + \mathrm{clip}\left(s_i^{\mathrm{des}} - s_i^{\ell}, -a_{m_i}^{\mathrm{dec}}\Delta\tau, a_{m_i}^{\mathrm{acc}}\Delta\tau\right), 0, V_{\max}^{m_i}\big), \\ \bar{\omega}_i^{\ell+1} &= \min\left(\omega_{\max}^{m_i}, \frac{s_i^{\ell+1}}{R_{\min}^{m_i}}\right), \\ \psi_i^{\ell+1} &= \mathrm{wrap}\big[\psi_i^{\ell} \\ &\quad + \mathrm{clip}\left(\mathrm{wrap}\left(\psi_i^{\mathrm{des}} - \psi_i^{\ell}\right), -\bar{\omega}_i^{\ell+1}\Delta\tau, \bar{\omega}_i^{\ell+1}\Delta\tau\right)\big], \end{aligned} \tag{5}$$

where $\Delta\tau = 60$ s. For the surface stage, $a_{\mathrm{s}}^{\mathrm{acc}} = 0.02$ m/s$^2$, $a_{\mathrm{s}}^{\mathrm{dec}} = 0.03$ m/s$^2$, $\omega_{\max}^{\mathrm{s}} = 1$°/s, and $R_{\min}^{\mathrm{s}} = 200$ m. For the underwater stage, $a_{\mathrm{u}}^{\mathrm{acc}} = 0.002$ m/s$^2$, $a_{\mathrm{u}}^{\mathrm{dec}} = 0.003$ m/s$^2$, $\omega_{\max}^{\mathrm{u}} = 2$°/s, and $R_{\min}^{\mathrm{u}} = 100$ m. The minimum-radius term is applied only while the vehicle is moving; a commanded zero action decelerates the vehicle without changing its course. The vehicle then advances over each substep using

$$\mathbf{x} \leftarrow \mathbf{x} + \left(\mathbf{q}_i^{\mathrm{prop}} + \mathbf{1}_{\{m_i = \mathrm{u}\}}\mathbf{u}_{\mathrm{ocean}}\right)\Delta\tau, \tag{6}$$

with the current held fixed over the 10 min control interval. Thus the surface desired command is a ground velocity, whereas the underwater desired command is a propulsion velocity before current advection. The speed is initialized to zero and the heading is initialized uniformly in $[0, 2\pi)$ at the beginning of each episode.

The motion model remains a horizontal mission-level point mass and does not resolve added mass, hydrodynamic drag, roll and pitch, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. A detailed six-degree-of-freedom model would require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The results therefore address mission-level planning and coordination rather than structural loading or low-level controller response.

## 2.3. Surface–Subsurface Two-Stage Task

The SUVs first perform surface tracking, using coordinated propulsion to follow the TC and approach its high-wind region. For the operational protocol, we assume that descent begins after the surface fleet enters the terminal region defined by a final tracking error of at most 200 km. The simulation uses this 200 km value as the upper bound on the tracking error at the surface-to-subsurface transition and initializes the subsurface vehicles within the corresponding

200 km disk around the TC center at the end time of the surface segment. Because the present simulation does not model descent dynamics, the surface and subsurface stages are represented as two time-aligned but state-independent episodes. The descent duration is simplified: after the surface stage, each vehicle is assumed to arrive immediately at one prescribed depth. Each depth is a separate configuration with four SUVs, and a single episode does not visit multiple depths. At the end of the underwater observation task, the operational concept returns the SUVs to the sea surface for recovery. This ascent and recovery phase is not modeled here; the paper models only the surface observation-and-tracking stage and the subsequent fixed-depth underwater observation stage.

The local observation of SUV $i$ is

$$\mathbf{o}_i(t) = \begin{cases} \left[\mathbf{s}_i(t),\ \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{s}, \\ \left[\mathbf{h}_i(t),\ \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{u}, \end{cases} \tag{7}$$

where $\mathbf{s}_i(t)$ is the stage-specific measurement block, $\mathbf{v}_i(t)$ is the normalized effective horizontal propulsion velocity $\mathbf{v}_i = \mathbf{q}_i^{\mathrm{prop}}/V_{\max}^{m_i(t)}$, and $\mathbf{h}_i(t)$ is an optional causal history block. In the final surface experiments, $\mathbf{s}_i(t) \in \mathbb{R}^{3N}$ contains the normalized speed and two wind components at 10 m above the sea surface sampled by all $N$ SUVs; the surface policy has no history block, and its effective ground velocity is augmented by $[\cos\psi_i, \sin\psi_i]$. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater active-control comparison, no inter-vehicle communication is assumed. The observation contains no separate instantaneous-current block in addition to the history; instead, define the local current feature $\mathbf{c}_{i,t} = [\|\mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,t}, z_i, t)\|, u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y}]^{\top}$, where $\mathbf{u}_{\mathrm{ocean}} = (u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y})$, and $\mathbf{h}_i(t) = [\mathbf{c}_{i,t-4}^{\top}, \dots, \mathbf{c}_{i,t}^{\top}]^{\top} \in \mathbb{R}^{15}$, which is the causal five-frame history of SUV $i$'s current speed and two components at its assigned depth; before five measurements are available, the earliest observed frame is repeated to fill the history. Each underwater SUV receives neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. These two communication assumptions are intentionally different and mark the bounds of the present study. The idealized global surface link represents a favorable satellite or radio-connected operating mode, which makes it an upper-bound assumption. The no-communication underwater case is the conservative opposite bound: shared policy parameters are loaded separately on each vehicle, but no observation, position, or action is exchanged at runtime. The surface payload is $3N$ scalar wind values per vehicle per decision, whereas the underwater inter-vehicle message payload is zero. The reported experiments do not model bandwidth, packet loss, acoustic latency, or intermittent surfacing. We do not present either the ideal surface link or the fully decentralized subsurface case as a validated communication system. The resulting surface observation dimension is $d_o^{\mathrm{s}} = 3N+2+2 = 3N+4$. The underwater five-frame history contains $5 \times 3 = 15$ scalars; adding the two normalized propulsion components and the heading representation $[\cos\psi_i, \sin\psi_i]$ gives $d_o^{\mathrm{u}} = 15 + 2 + 2 = 19$.

During training, the reward function can access the distance between each SUV and the best-track TC center. This distance supervises the association between local 10 m wind patterns and proximity to the TC center/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The trained policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{8}$$

and is projected onto the unit disk before being mapped to a desired velocity command in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}^{m_i(t)} \bar{\mathbf{a}}_i(t). \tag{9}$$

This mapping ensures that the desired and effective speeds never exceed the mode-specific speed limit. Here $\bar{\mathbf{a}}_i(t)$ is the executed normalized command after action smoothing, unit-disk projection, and any stage-specific safety projection; without a safety intervention, $\bar{\mathbf{a}}_i(t) = \tilde{\mathbf{a}}_i(t)$. In the surface stage, the action specifies a desired ground velocity. In the underwater stage, the same unit-disk command specifies a desired propulsion speed and course. In both stages, the effective speed and heading evolve according to the dynamics defined above.

The cooperative task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\Omega_i\}, O, \gamma \rangle, \tag{10}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{A}_i$ and $\Omega_i$ are the local action and observation spaces of vehicle $i$, $P(s_{t+1} \mid s_t, \mathbf{a}_t)$ is the transition probability, $R(s_t, \mathbf{a}_t, s_{t+1}) \in \mathbb{R}^N$ is the local-reward vector, and $O(\mathbf{o}_{t+1} \mid s_{t+1}, \mathbf{a}_t)$ is the joint observation function. The shared team reward before clipping is $\bar{r}_t = N^{-1} \sum_{i=1}^{N} r_{i,t}$, where $r_{i,t}$ is the $i$th element of $R(s_t, \mathbf{a}_t, s_{t+1})$, and $\gamma$ is the discount factor.

At each 10 min interval, the policy forms a closed perception–decision–action loop and does not follow a precomputed route:

$$\begin{aligned} \mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}. \end{aligned} \tag{11}$$

The updated observation becomes the next control input. This feedback structure lets the fleet adapt to changing wind and current fields, partial observability, and deployment uncertainty without a static waypoint sequence.

**Table 1**
Environmental datasets used to construct the 500-TC observation environments. The dataset integrates ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

| Data product | Source | Variables used | Native grid | Native time |
|---|---|---|---|---|
| Surface wind at 10 m above sea level | ERA5 | $u_{10}$, $v_{10}$, speed, direction | $0.25° \times 0.25°$ | Hourly |
| Current at 0–60 m ocean depth | HYCOM | `water_u`, `water_v`, speed at 0/10/20/30/40/50/60 m | $\sim 0.04°$ lat $\times 0.08°$ lon | 3-hourly |
| Best track | IBTrACS | Position, time, wind, pressure | Point track | Mainly 3/6-hourly |

Note: although the SUV platform can carry sensors for observing temperature, salinity, and currents, the reinforcement-learning policy uses only surface wind and subsurface current observations as environmental inputs.

# 3. Data and Simulation Environment

## 3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [60]. Wind speed and direction are calculated from $u_{10}$ and $v_{10}$. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [61]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [62]. The dataset-selection domain used to identify candidate systems is 98.7°E–171.6°E and 6.8°N–56.7°N. Individual event fields are stored on event-specific crops, so a track may extend beyond this screening box. The overall event set spans 19 November 2007 to 3 December 2025 UTC. The surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by `water_u` and `water_v`, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the event-specific environments.

## 3.2. Spatiotemporal Alignment and Track Segmentation

Hourly ERA5 winds and three-hourly HYCOM currents are linearly interpolated to a common 10 min simulation clock and queried at each SUV position. The environmental trajectories are precomputed and do not respond to vehicle motion; this one-way forcing assumption is appropriate for mission-level planning at the scale of a TC. The policy receives only local reconstructed values at the current time, not future fields or the complete reanalysis map. Invalid interpolated values are replaced by a neutral value, and all policy observations are checked for finiteness. The best-track center is used only for training rewards and evaluation.

Commands are held constant within each decision interval:

$$\mathbf{a}_i(t) = \mathbf{a}_{i,k}, \quad t_k \le t < t_{k+1}, \quad t_{k+1} - t_k = \Delta t = 10\,\text{min}. \tag{12}$$

The environmental fields, TC track, action, and position update use the same causal time grid. The 10 min step balances control responsiveness against the hourly and three-hourly resolution of the source products; a shorter step would add samples without adding resolved information. The same interval is used for IPPO and all traditional planners.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \tag{13}$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are longitudinal and latitudinal increments in degrees, and 111 km is the mean meridional arc length per degree. Target segment lengths are sampled uniformly from 300–500 km, with a fixed 250 km stride, 50–250 km overlap, and an average overlap of about 150 km. Residual track shorter than 300 km is discarded. After assignment to 10 min track samples, realized lengths range from 292.1 to 507.1 km, with a median of 396.4 km.

The 500-TC dataset yields 7,248 segments from 495 TCs. The 450-TC training split contributes 6,480 segments from 445 usable TCs, whereas the 50 held-out TCs contribute 768 segments. The split is made at the TC level, so no TC appears in both sets. Segment durations range from 2.2 to 103 h. The median duration was 16.5 h and the mean was 20.0 h. These segments provide mission-scale evaluation windows; battery depletion is not simulated. The platform can carry current, CTD, pH, chlorophyll, and turbidity sensors [28], while the control experiments use current velocity as the decision-making observation.

# 4. Multi-Agent Reinforcement Learning Method

## 4.1. Blue Whale-Class Platform Modeling

Each SUV is represented as a horizontal point mass with mode-dependent speed bounds and position updates from Eqs. (1)–(2). The surface actor uses globally shared 10 m wind observations and its normalized velocity, whereas

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[1] ERA5 download portal: `https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels`.

[2] HYCOM GLBy0.08 dataset: `https://www.hycom.org/data/glby0p08/expt-93pt0`; HYCOM ESPC-D-V02 dataset: `https://www.hycom.org/data/espcd-v02`.

[3] IBTrACS download portal: `https://www.ncei.noaa.gov/products/international-best-track-archive`.

**Table 2**
Public Blue Whale reference and the two simulation configurations. The second simulation column is a simple parameterization of the present-day reference platform, whereas the third represents a possible future-upgrade configuration with higher surface and subsurface speeds for comparison. These scenarios are modeling assumptions, not certified vehicle data. The 20 h value is a common simulation horizon, not a platform-endurance claim.

| Property | Public reference | Current-platform scenario | Future-upgrade scenario |
|---|---|---|---|
| Role | Engineering reference | Simple current-platform parameterization | Potential future capability |
| Surface speed | 23 kn, ≈11.83 m/s | 12 m/s | 20 m/s |
| Submerged speed | 3 kn, ≈1.54 m/s | 1.5 m/s | 2.0 m/s |
| Public-reference submerged endurance | ≥4 h at 3 kn; ≥72 h standby | – | – |
| Underwater evaluation horizon | – | 20 h simulation | 20 h simulation |
| Sampling depth | ≈60 m | 0–60 m, seven layers | 0–60 m, seven layers |
| Payload | ≥800 kg | 800 kg | 800 kg |
| Surface fleet size | Single platform | 4/8/12/16 | 4/8/12/16 |
| Decision interval | Not specified | 10 min | 10 min |
| Surface formation | – | 100 km ring, 360°/$N$ | 100 km ring, 360°/$N$ |

**Table 3**
IBTrACS TC translation speeds for the 500-TC dataset. The percentages denote the fraction of main-track samples not exceeding each platform speed. These values describe the speed envelope only; tracking also depends on bearing error and remaining segment duration.

| Quantity | P50 | P90 | P95 | P99 | Samples | TCs |
|---|---|---|---|---|---|---|
| TC translation speed (m s$^{-1}$) | 5.14 | 9.77 | 13.38 | 20.06 | 32,352 | 500 |

| Envelope | Samples within speed | Samples above speed | Interpretation |
|---|---|---|---|
| 1.5 m/s submerged | 3.7% | 96.3% | local sampling, not TC chasing |
| 2.0 m/s submerged | 7.8% | 92.2% | local sampling, not TC chasing |
| 12 m/s surface | 93.8% | 6.2% | current-platform scenario |
| 20 m/s surface | 98.9% | 1.1% | future-upgrade scenario |

the underwater actor uses a causal five-frame history of its local current and normalized velocity. The surface control variable is defined in the Earth-fixed frame as a desired ground velocity. Therefore, ocean-current compensation is not included as a separate term in the surface position update. Underwater motion retains advection because 1.5–2.0 m/s propulsion is comparable to the TC-induced current.

The two simulation configurations compare present and potential future platform capabilities. The 12 m/s surface and 1.5 m/s submerged limits are close to the public Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent a future-upgrade case. Table 2 reports the complete parameterization and distinguishes modeling assumptions from certified specifications.

Across 32,352 IBTrACS track samples, the median, 95th-percentile, and 99th-percentile TC translation speeds are 5.14, 13.38, and 20.06 m/s (Table 3). The 12 and 20 m/s surface limits cover 93.8% and 98.9% of these samples, respectively. The submerged limits cover only 3.7% and 7.8200 km initialization region and does not attempt to overtake and follow the TC center.

## 4.2. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region, with the final tracking error as the primary operational metric. Let $d_{i,k+1}$ denote the post-transition distance from SUV $i$ to the TC center after executing the command at decision step $k$. The surface reward used in the implementation is

$$\begin{aligned} r_i^{\mathrm{surf}} = & -\frac{\min(d_{i,k+1}, D_{\max})}{D_{\mathrm{scale}}} + b_{\mathrm{near}} \mathbf{1}_{d_{i,k+1} < d_{\mathrm{near}}} \\ & -\frac{W_{\mathrm{form}} \max(0, d_{i,k+1} - R_{\mathrm{form}})}{D_{\mathrm{scale}}} \\ & - \lambda_u \|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2 + r_i^{\mathrm{coast}}, \end{aligned} \tag{14}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. Here $D_{\max} = 300$ km caps the proximity cost, $D_{\mathrm{scale}} = 100$ km normalizes the distance terms, and $d_{\mathrm{near}} = 100$ km defines the near-center reward region. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term $r_i^{\mathrm{coast}}$ represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their primary objective is not to follow the surface TC center directly, but to observe the highest possible current speed. Nominal and current-weighted spatial coverage are then evaluated as complementary measures of observation extent and observation value. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For the active-control comparison, the current-seeking reward is

$$\begin{aligned} r_i^{\text{sub}} = \min&\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\text{norm}}}, R_{\max}\right) \\ &- \lambda_a \|\tilde{\mathbf{a}}_{i,k}\|_2^2 + \beta_{\|} \max(0, \cos\theta_i) \\ &+ r_i^{\text{coast}}, \end{aligned} \tag{15}$$

where $V_{\text{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_a = 0.05$, and $\beta_{\|} = 0.5$. The term $-\lambda_a \|\tilde{\mathbf{a}}_{i,k}\|_2^2$ penalizes large normalized actions. The angle $\theta_i$ denotes the direction difference between the effective propulsion velocity and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, we define $\cos\theta_i = 0$ when either vector is zero and otherwise set $\cos\theta_i = (\mathbf{q}_{i,k}^{\text{prop}} \cdot \mathbf{u}_{i,k})/(\|\mathbf{q}_{i,k}^{\text{prop}}\|_2 \|\mathbf{u}_{i,k}\|_2)$. The current-intensity reward is evaluated at $\mathbf{x}_{i,k}$ before the state update. For the local maximum-current objective, passive drift is retained as the no-propulsion reference because it follows the ambient flow directly and does not depend on a learned reward policy.

### 4.3. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance $D_{\text{buf}}$ receives

$$r_i^{\text{coast}} = \begin{cases} -P_{\text{land}}, & D_i = 0, \\ -W_b \max(0, 1 - D_i/D_{\text{buf}}), & 0 < D_i < D_{\text{buf}}, \\ 0, & D_i \geq D_{\text{buf}}, \end{cases} \tag{16}$$

where $D_i = D(\mathbf{x}_{i,k+1})$ is the post-transition offshore distance, with $D_i = 0$ on land, and $P_{\text{land}}$ and $W_b$ are positive penalty coefficients.

For the surface stage, three complementary mechanisms reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The main surface experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. Best-of-rejection deployment favors offshore initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. For the subsurface stage, the nearshore penalty is retained and an episode is truncated at landfall, but offshore-only initialization and action-level safety filtering are not applied. These mechanisms reduce land exposure in the surface results but do not guarantee offshore operation for every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_k^{\text{filt}} = (1 - \beta)\mathbf{a}_k^{\text{raw}} + \beta \mathbf{a}_{k-1}^{\text{filt}}, \tag{17}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions, which prevents future-information leakage. The filtered command $\mathbf{a}_k^{\text{filt}}$ is then passed through the unit-disk projection and speed mapping in Eq. (9).

### 4.4. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. This return defines a fully cooperative task. Its reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes the cooperative and parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. Actions are sampled independently, and the per-agent values are summed for training without a global-state centralized critic. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward, with value estimate $V_\phi(\mathbf{o}_t) = \sum_{i \in \mathcal{I}} v_\phi(\mathbf{o}_{i,t})$, and a single clipped PPO objective is optimized over the factorized multi-agent action vector:

$$\begin{aligned} L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(&\rho_t(\theta)\hat{A}_t, \\ &\text{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right) \hat{A}_t\Big)\Big], \end{aligned} \tag{18}$$

where

$$\rho_t(\theta) = \prod_{i \in \mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t} \mid o_{i,t})}, \tag{19}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. During execution, each SUV computes its action from its own observation and the shared policy network.

To compare reinforcement-learning algorithm choices, IPPO, SAC, and TD3 were trained for the same $2 \times 10^6$ environment steps on representative surface and subsurface cases. In the surface comparison, IPPO achieved a final error of 312.0 km, compared with 510.0 km for SAC and 526.2 km for TD3. In the subsurface comparison, IPPO obtained an episode-level current-weighted coverage $C_w$ of

**Algorithm 1** Training and deployment of parameter-sharing IPPO

```
Input: Training TC split, stage-specific environment, fleet size $N$, speed limit $V_{\max}$, reward coefficients, and random seed
Output: Shared actor and critic parameters
Initialize the actor, critic, and parallel environment workers
for training iteration $m = 1, \dots, M$ do
    Reset the stage-specific environment and obtain $\mathbf{o}_{i,t}$ for all $i = 1, \dots, N$
    for rollout step $t$ do
        Sample or select an action for each SUV from $\pi_\theta(\cdot \mid \mathbf{o}_{i,t})$
        Apply the stage-specific action filter, project each agent's action onto the unit disk, and apply the safety filter
        Advance all SUVs and collect the team-average reward
    end for
    Compute GAE advantages and returns for the team-average reward
    Update the shared actor and critic with the clipped PPO objective
end for
Freeze the actor and load the same parameters on every SUV
for each deployment episode do
    Observe local or shared data according to the stage communication model
    Compute $\mathbf{a}_{i,t}$ from the shared actor
    Apply action smoothing, circular projection, and safety filtering
    Execute the action and observe the next state
end for
```

1532.7, compared with 1211.6 for SAC and 1239.8 for TD3. IPPO also required no replay buffer in these representative runs. We use IPPO for the main, formation, and planner comparisons and treat this algorithm comparison as bounded evidence without claiming a universal ranking.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix A. The project repository and software package sources are listed in the Code and package availability section.

## 4.5. Surface and Subsurface Evaluation

The center-distance reward is used only during training and is absent from validation. Validation tests whether a trained policy can infer the TC direction from wind observations in unseen events. The primary surface metric is the final fleet-averaged SUV-to-TC-center distance. A smaller value indicates closer core entry and higher observation benefit. Time-averaged distance is reported as a secondary process diagnostic.

For training seed $s$ and its validation dataset, the seed-level value is

$$D^{(s)}_{\text{final}} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e}(T_e) - \mathbf{x}_{\text{TC},e}(T_e) \right\|_2, \quad (20)$$

where $T_e$ is the final decision time of validation segment $e$, and $M_s$ is the number of validation segments. Results are averaged over ten training seeds and reported with the sample standard deviation.

The process-level fleet-mean time-averaged distance for the same seed is

$$D^{(s)}_{\text{avg}} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{K_e} \sum_{k=1}^{K_e} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e,k} - \mathbf{x}_{\text{TC},e,k} \right\|_2, \quad (21)$$

where $K_e$ indexes the post-transition decision intervals. Distance is averaged over SUVs, episode time, and validation segments, then over training seeds.

For the execution-filter comparison, let $L_{i,e}$ be the path length of SUV $i$ in validation segment $e$, computed from consecutive recorded positions, and let $T_e$ be the corresponding episode duration. The reported fleet-mean ground speed is

$$\bar{v}^{(s)} = \frac{1}{M_s} \sum_{e=1}^{M_s} \left[ \frac{1000}{N\, T_e} \sum_{i=1}^{N} L_{i,e} \right], \quad (22)$$

where $L_{i,e}$ is expressed in km, $T_e$ in seconds, and the factor 1000 converts km to m. The reported maximum speed is the mean over validation segments of each segment's maximum simulated propulsion speed:

$$V^{(s)}_{\text{peak}} = \frac{1}{M_s} \sum_{e=1}^{M_s} \max_{k=0,\dots,K_e-1} \; \max_{i=1,\dots,N} s_{i,e,k}, \quad (23)$$

where $s_{i,e,k}$ is the effective propulsion speed. Because the final surface configuration applies no ocean-current term, this speed is also the simulated ground speed.

Surface evaluation covers 4/8/12/16 SUVs on a 100 km ring with equal angular spacing. Training samples formation-center offsets within an 800 km radius; the trained policies are evaluated at 100, 200, 400, 600, and 800 km. These are mission-scale test conditions, not tuned deployment optima. Metrics are first averaged over validation segments within each seed, then summarized over ten seeds. Paired 12-versus-20 m/s tests are two-sided and uncorrected across the twenty fleet-size/radius comparisons, so their $p$-values are exploratory.

For the subsurface evaluation, each depth from 0, 10, 20, 30, 40, 50, and 60 m is tested as a separate 20 h configuration with four SUVs and 1.5 or 2.0 m/s speed limits. The 200 km terminal surface tracking error marks the assumed start of descent and is used as the upper bound of the subsurface initialization radius at the corresponding surface-segment end time. The descent time is neglected: each SUV is initialized directly at its assigned depth. The surface endpoint positions themselves are not transferred. There is no inter-vehicle communication, so each SUV uses only its own five-frame current history and propulsion state. Active IPPO, random control, and passive drift share the same episodes, deployment seeds, motion model, and speed bound.

The three strategies differ only in how propulsion is generated:

$$\begin{aligned} \mathbf{a}_i^{\text{active}} &= \operatorname{clip}\left(\pi_\theta(\mathbf{o}_i), -1, 1\right), \\ \mathbf{a}_i^{\text{random}} &\sim \mathcal{U}\left([-1,1]^2\right), \\ \mathbf{a}_i^{\text{drift}} &= \mathbf{0}. \end{aligned} \tag{24}$$

During evaluation, the active IPPO policy uses its deterministic mean action, whereas random control resamples a two-dimensional action every 10 min. Active IPPO and random control share the same propulsion limit and equations of motion; only the active policy conditions its action on the local current history. Passive drift applies zero propulsion and advances solely by ambient current advection.

For evaluation reporting, let $K$ be the number of recorded position increments and let $k = 0, \dots, K$ index the archived positions. The mean experienced current speed and nominal coverage are

$$\bar{V}_i(z) = \frac{1}{K+1} \sum_{k=0}^{K} \left\| \mathbf{u}_{\text{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2 . \tag{25}$$

$$C(z) = A_{\text{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K} \mathcal{G}(\mathbf{x}_{i,k}) \right|, \tag{26}$$

where $\mathcal{G}(\mathbf{x})$ is the 0.25° grid cell containing position $\mathbf{x}$, $|\cdot|$ counts unique cells, and $A_{\text{cell}} \approx (0.25^\circ \times 111\ \text{km})^2$ is the nominal grid-cell area used for cross-method comparison. The figures report the fleet mean $N^{-1} \sum_{i=1}^{N} \bar{V}_i(z)$ and the nominal coverage $C(z)$. Because neither current intensity nor coverage alone captures observation value, we also report the episode-level current-weighted coverage

$$C_{\text{w}}(z) = \frac{1}{M} \sum_{e=1}^{M} \left[ \frac{1}{N} \sum_{i=1}^{N} \bar{V}_{i,e}(z) \right] C_e(z), \tag{27}$$

where $M$ is the number of paired validation episodes. This quantity assigns a higher value to covering a broad region while encountering a strong current within that region, without assuming that either objective dominates in every episode.

### 4.6. Traditional Planning Baselines

We include four non-learned planners as secondary references [36, 38, 40]. The artificial-potential-field planner moves along the local gradient of observed wind or current speed and adds an offshore repulsion inside the coastal buffer. The receding-horizon A* planner builds an online local value map, re-plans a three-step minimum-cost route, executes the first action, and refreshes the plan every 15 decisions. The boustrophedon lawnmower planner follows fixed parallel coverage legs with phase offsets between vehicles and uses no environmental value map. The information-greedy planner evaluates eight candidate unit actions and selects the one with the largest predicted local wind or current value plus the coastal preference.

None of the planners receives the TC center, the complete reanalysis field, future frames, or the learned policy state. Surface planners use the same shared wind samples and fleet navigation state as IPPO. Underwater, each vehicle maintains a private map because communication is disabled. All four use the same 10 min interval, speed envelope, circular projection, surface safety filter, and subsurface landfall termination. They require no training and have no training-seed variance. Surface planners are evaluated on all 768 held-out segments. Underwater planners use the same 500 paired episodes per depth-speed setting. These episodes comprise 50 test TCs with ten deployment seeds each.

## 5. Simulation and Results

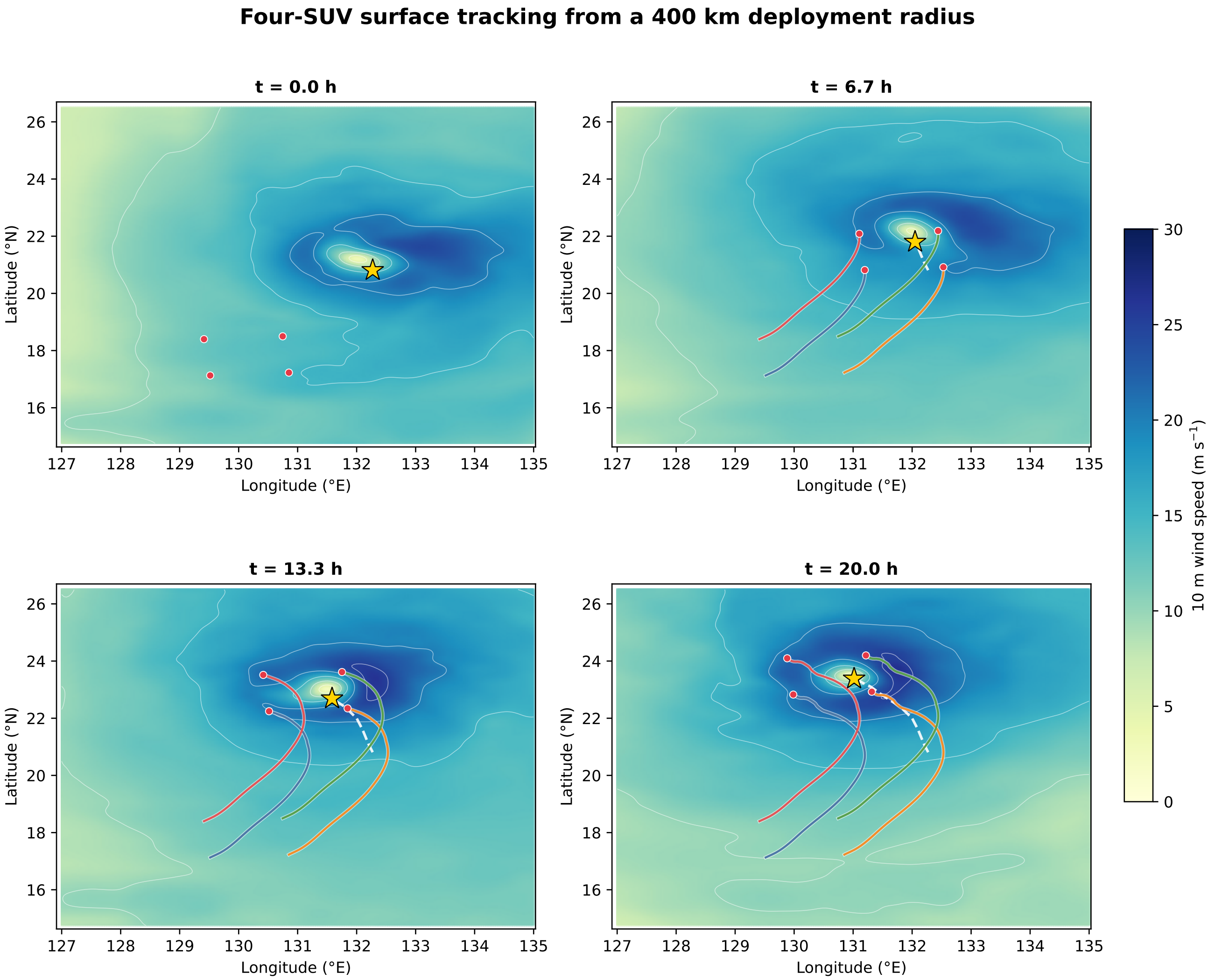
### 5.1. Surface Active Tracking

Using the evaluation protocol in Section 4.5, we evaluate the trained surface policies across four fleet sizes, two speed limits, five deployment radii, and ten independent training seeds. Final tracking error in Eq. (20) is the primary metric. Time-averaged distance is a secondary diagnostic of how closely the fleet remains to the moving TC, while path length and mean ground speed characterize motion demand.

Figure 4 reports the population-level speed-limit comparison. Averaged over fleet sizes and training seeds, the 20 m/s policy obtains final errors of 204.8, 217.2, 272.5, 379.3, and 540.8 km at deployment radii of 100, 200, 400, 600, and 800 km. The corresponding 12 m/s values are 236.7, 255.1, 335.5, 468.4, and 651.3 km. Thus, 20 m/s gives the lower final error in all 20 matched fleet-size/radius comparisons, and its fleet-averaged advantage grows from 31.9 km at 100 km to 110.5 km at 800 km. Both policies remain farther from the center as deployment error increases, because finite speed and episode duration cannot remove every initial displacement.

The time-averaged distance shows the same ordering: 20 m/s has the smaller distance in all 20 matched comparisons, with a median 12-minus-20 m/s paired difference of 49.7 km. The two-sided paired tests give $p < 0.05$ in every case; these tests are exploratory because they are not corrected across the 20 comparisons. The final-error benefit is larger than the time-averaged benefit at both ends of the deployment range, indicating that the higher-speed policy continues reducing residual separation late in the episode.

The 12 m/s policy reduces path length per SUV by 83.4–220.1 km across the five radii and lowers mean ground speed



Four-SUV surface tracking from a 400 km deployment radius

t = 0.0 h

t = 6.7 h

t = 13.3 h

t = 20.0 h

**Figure 3:** Four-SUV surface-tracking process for KHANUN (2023) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

by 1.75–3.93 m/s, whereas the 20 m/s policy achieves closer final tracking.

The representative trajectories clarify how the aggregate result is produced. Fig. 3 follows the four-SUV, 400 km KHANUN case. The vehicles first turn along the observed wind structure and then reduce their separation from the center; by 19.5 h they have entered the high-wind branch. Individual trajectories differ because each SUV has its own state, but the fleet-level motion follows a consistent inward correction.

Figure 5 separates the effects of fleet size and deployment error. At 100–200 km, the formations converge rapidly and stay concentrated near the TC core. At 600–800 km, trajectories lengthen and fan out before turning inward, and final positions remain farther from the center. Increasing the fleet from 4 to 16 SUVs adds sample branches and broadens the approach region, but it does not replace the long-range transit. In the representative 12- and 16-SUV cases, paths overlap near the core once the shared wind observations guide the fleet toward the same high-wind region. Deployment error therefore controls the required transit distance, whereas fleet size mainly controls how broadly the approach is sampled.

## 5.2. Surface Formation Comparison

The ring, line, and grid formations provide three different ways to allocate spatial observations across eight SUVs. Each was trained with the same 2-million-step protocol, deployment distribution, and safety layer at both 12 and 20 m/s, then evaluated without retraining at 100–800 km (Table 4). Averaged over the ten speed–radius conditions, the line gives the lowest mean final error of 320.8 km, followed by the ring at 341.2 km and the grid at 396.0 km. The line is best in eight of the ten speed–radius conditions; the ring is best at 600 and 800 km for 12 m/s.

Formation geometry is summarized at each post-transition state $\mathbf{x}_{i,e,k}$. With centroid $\bar{\mathbf{x}}_{e,k} = N^{-1} \sum_i \mathbf{x}_{i,e,k}$,

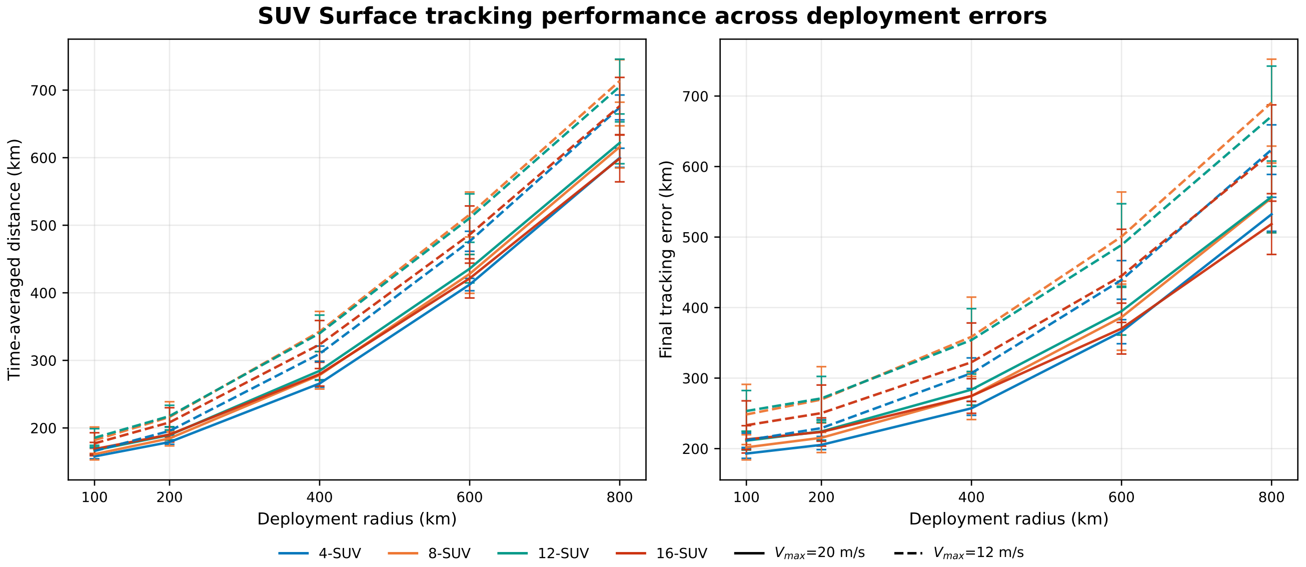


**Figure 4:** SUV surface tracking performance across deployment errors. The left panel shows the fleet-mean time-averaged distance to the TC center, representing whether the fleet remains close to the moving TC throughout an episode. The right panel shows the final tracking error to the TC center at the end of the episode; smaller values indicate higher operational observation benefit. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs using ten random seeds. Error bars show the sample standard deviation.

the episode means are

$$R_{\text{form},e} = \frac{1}{K_e}\sum_{k=1}^{K_e}\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_{i,e,k} - \bar{\mathbf{x}}_{e,k}\right\|_2, \tag{28}$$

$$S_{\text{pair},e} = \frac{1}{K_e}\sum_{k=1}^{K_e}\frac{2}{N(N-1)}\sum_{1\le i<j\le N}\left\|\mathbf{x}_{i,e,k} - \mathbf{x}_{j,e,k}\right\|_2. \tag{29}$$

Reported formation radii and pairwise spacings are means of $R_{\text{form},e}$ and $S_{\text{pair},e}$ over the 768 validation segments at each speed–radius condition.

The grid produces the shortest per-vehicle path, averaging 392.7 km, followed by the ring at 416.5 km and the line at 482.6 km. The line contracts the mean formation radius to 57.9 km and the pairwise spacing to 86.3 km. The ring retains a 99.3 km radius and 142.6 km spacing, giving a broader observation footprint. The grid preserves the widest footprint, with a 103.1 km radius and 156.1 km spacing, but has the largest final tracking error. We retain the ring for the main broad-coverage configuration. The line favors tighter tracking at most radii, and the grid favors compact trajectories and a wide footprint; none is treated as a universally superior formation.

Figure B.1 visualizes one matched KHANUN segment. The formation geometry is not rigidly preserved because each policy moves under the same shared wind observation but with its own state. This representative episode illustrates how the compact line formation remains spatially concentrated, whereas the ring and grid retain a broader sampling footprint. Formation choice is based on the aggregate 768-segment results; the selected episode is illustrative. On this basis, we retain the ring as the main configuration and report the alternatives as a sensitivity study without claiming that one formation is universally optimal.

## 5.3. Execution-Layer Action Filtering

Table 5 reports the low-pass filtering ablation over the same trained actor, validation dataset, and 10 min decision interval. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1019.2–1160.0 km per vehicle, with mean speeds of 14.85–16.45 m/s and maximum speeds of 19.46–19.47 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 497.4–670.8 km, mean speed to 8.49–11.01 m/s, and maximum speed to 14.25–17.97 m/s. The final tracking error changes from 355.4 to 251.3 km for four SUVs, from 324.7 to 277.8 km for eight SUVs, from 312.0 to 278.8 km for 12 SUVs, and from 260.2 to 256.1 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 313.1 to 266.0 km, a reduction of 47.1 km or 15.0%. Filtering improves final tracking for every fleet size.

The same comparison shows reductions for every fleet size: path length decreases by 651.7, 614.9, 521.9, and 459.1 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 5.22, 4.14, 3.50, and 1.49 m/s. The execution filter improves final tracking while substantially shortening trajectories and reducing mean and peak speed.

## 5.4. Subsurface Strategy Comparison

For each depth, four SUVs seek higher current speed; the central comparison is the fleet-mean current speed experienced along each trajectory, together with nominal and current-weighted spatial coverage. The comparison follows the protocol in Section 4.5. The final ten-seed results cover

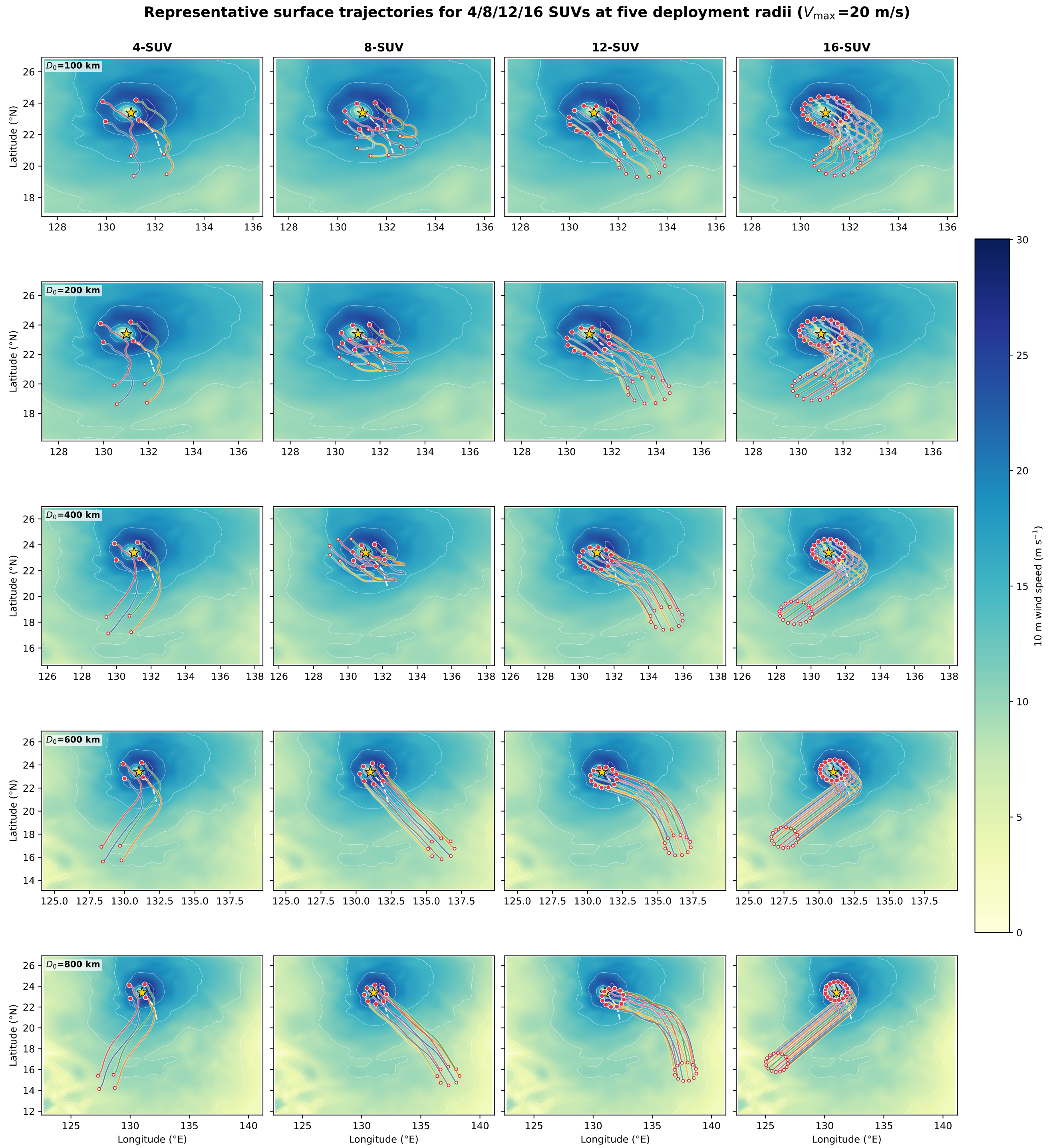

Representative surface trajectories for 4/8/12/16 SUVs at five deployment radii ($V_{\max}$=20 m/s)
4-SUV
8-SUV
12-SUV
16-SUV
$D_0$=100 km
$D_0$=200 km
$D_0$=400 km
$D_0$=600 km
$D_0$=800 km
Latitude (°N)
Longitude (°E)
10 m wind speed (m s$^{-1}$)
0
5
10
15
20
25
30

**Figure 5:** Representative KHANUN (2023) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

all seven depths from 0 to 60 m. At every depth and both propulsion limits, active IPPO attains a higher mean experienced current speed than passive drift and random control. The configuration-mean improvement over passive drift ranges from 2.2% to 4.6%, while the corresponding improvement over random control ranges from 2.3% to 4.6%.

Active IPPO provides an intended observation advantage. Relative to passive drift, its nominal coverage increases by 3.09–5.68 times and its current-weighted coverage increases by 2.87–4.99 times. Across the 140 training-seed comparisons, active control has a higher mean experienced current speed in 126 cases. False-discovery-rate-adjusted

**Table 4**

Surface initial-formation comparison for eight SUVs. Each policy was trained with formation-center offsets sampled inside an 800 km radius and evaluated without retraining at fixed offsets of 100, 200, 400, 600, and 800 km on the same 768 held-out TC segments. Error is the final SUV-to-best-track-center distance and path is the per-vehicle trajectory length. Bold entries are the best value in each speed/radius row; error and path are assessed separately. All distances are in km.

| Speed | Train radius | Eval radius | Ring | | Line | | Grid | |
|---|---|---|---|---|---|---|---|---|
| (m/s) | (km) | (km) | Error | Path | Error | Path | Error | Path |
| 12 | 800 | 100 | 210.4 | 284.4 | **184.8** | 330.4 | 295.8 | **219.3** |
| 12 | 800 | 200 | 228.6 | 317.2 | **202.4** | 364.5 | 317.7 | **241.2** |
| 12 | 800 | 400 | 312.0 | 364.2 | **295.7** | 409.9 | 408.5 | **272.8** |
| 12 | 800 | 600 | **445.9** | 383.9 | 452.6 | 408.5 | 542.6 | **291.4** |
| 12 | 800 | 800 | **636.5** | 372.4 | 655.6 | 394.1 | 733.5 | **285.9** |
| 20 | 800 | 100 | 197.5 | **356.2** | **156.4** | 435.3 | 204.7 | 383.2 |
| 20 | 800 | 200 | 209.9 | **404.6** | **168.0** | 489.5 | 217.7 | 433.7 |
| 20 | 800 | 400 | 266.6 | **502.5** | **228.4** | 599.5 | 277.7 | 537.7 |
| 20 | 800 | 600 | 371.1 | **571.9** | **342.1** | 676.6 | 390.6 | 616.4 |
| 20 | 800 | 800 | 533.9 | **607.7** | **522.0** | 717.2 | 571.1 | 645.5 |

**Table 5**

Execution-layer low-pass filtering for 4/8/12/16 surface SUVs on 768 held-out segments from 50 unseen TCs in the 500-TC dataset. Each trained actor is evaluated with the filter off and on ($\beta = 0.8$) on the same validation episodes.

| Fleet | Filter | Final tracking error (km) | Path per SUV (km) | Mean speed (m/s) | Maximum speed (m/s) |
|---|---|---|---|---|---|
| 4 | Off | 355.4 | 1160.0 | 16.45 | 19.47 |
| 4 | On | **251.3** | **508.2** | **8.49** | **14.25** |
| 8 | Off | 324.7 | 1119.5 | 16.11 | 19.47 |
| 8 | On | **277.8** | **504.7** | **8.65** | **15.32** |
| 12 | Off | 312.0 | 1019.2 | 14.85 | 19.46 |
| 12 | On | **278.8** | **497.4** | **8.52** | **15.96** |
| 16 | Off | 260.2 | 1129.9 | 16.29 | 19.47 |
| 16 | On | **256.1** | **670.8** | **11.01** | **17.97** |

Final tracking error is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off and on use the same trained actor.

paired tests identify 114 significant improvements and six significant deficits relative to passive drift. Both nominal and current-weighted coverage improve significantly in all 140 comparisons. The current-speed gain is therefore modest and not universally positive at the training-seed level, but active control consistently increases the spatial extent and current-weighted value of the collected underwater data. Passive drift remains the low-motion reference and cannot deliberately relocate toward higher-current regions.

The propulsion limit consistently affects observation extent. At all seven depths, the 2.0 m/s policy obtains 1.18–1.34 times larger nominal coverage than the 1.5 m/s policy because the larger speed envelope permits more movement within the same 20 h episode. Under this fixed horizon, the 2.0 m/s setting therefore samples more locations and covers a larger part of the TC-induced current field. Mean experienced drift current decreases from 0.521 to 0.292 m/s between 0 and 60 m.

The reanalysis current fields contain pronounced horizontal gradients and localized high-speed bands. The fleet-mean experienced current speed in the final experiments ranges from 0.29 to 0.53 m/s across methods and depths, but this average includes the full sampled trajectory and does not isolate the strongest-current patches. Local current speeds can be substantially higher than the fleet mean and are of comparable magnitude to the 1.5 and 2.0 m/s propulsion limits. Active propulsion can relocate a vehicle toward such patches, whereas passive drift follows the ambient flow without steering. Active control uses this capability to obtain both a modestly higher mean current speed and a substantially larger observation footprint and data volume.

## 5.5. Comparison with Traditional Planners

Table 6 summarizes the comparison with four traditional planners. Across all 40 matched surface configurations, parameter-sharing IPPO attains the lowest mean final tracking error and time-averaged distance. Relative to the potential-field, A*, lawnmower, and information-greedy planners, its mean final error is reduced by 57.3, 101.0, 198.6, and 110.0 km, respectively. IPPO has the lower final error in 35 of 40 comparisons against potential field and in all 40 comparisons against each other planner. Its path is shorter than those of potential field and A* in all 40 comparisons. It is also shorter than the lawnmower and information-greedy paths in 26 and 34 of the 40 comparisons; their fleet-average paths are still longer, but the path advantage is less uniform. This result indicates that the learned controller preserves direct inward motion while adapting more effectively to local wind structure than the local-gradient and rolling-map planners.

**Table 6**
Comparison with four traditional planning baselines on the held-out TC segments from the same 500-TC train/test split. Surface values are means over the 4/8/12/16-SUV fleets, both speed limits, and all five deployment radii. Subsurface values are means over all seven depths and both propulsion limits. For surface tracking, lower errors and paths are better. For subsurface observation, higher current and coverage are better.

**Panel A: surface tracking**

| Method | Final error (km) | Time-averaged distance (km) | Path per SUV (km) |
|---|---|---|---|
| Parameter-sharing IPPO | **356.2** | **356.6** | **462.3** |
| Potential field | 413.4 | 404.2 | 728.8 |
| Receding-horizon A* | 457.2 | 429.4 | 823.9 |
| Lawnmower coverage | 554.7 | 478.4 | 509.1 |
| Information greedy | 466.2 | 429.4 | 528.7 |

**Panel B: subsurface observation**

| Method | Mean current (m/s) | Nominal coverage (km$^2$) | Current-weighted coverage ($10^3$ m s$^{-1}$ km$^2$) |
|---|---|---|---|
| Parameter-sharing IPPO | 0.408 | **3506** | **1.599** |
| Potential field | **0.452** | 2048 | 1.050 |
| Receding-horizon A* | 0.427 | 2310 | 1.097 |
| Lawnmower coverage | 0.408 | 2914 | 1.286 |
| Information greedy | 0.429 | 1549 | 0.739 |

Underwater, the four baselines are not dominated on every metric. The potential-field planner attains the highest mean experienced current at 0.452 m/s, followed by information-greedy control at 0.429 m/s and A* at 0.427 m/s. IPPO and lawnmower coverage are close at approximately 0.408 m/s. IPPO produces the largest nominal coverage of 3506 km$^2$. The corresponding nominal coverages are 2048 km$^2$ for potential field, 2310 km$^2$ for A*, 2914 km$^2$ for lawnmower coverage, and 1549 km$^2$ for information-greedy control. The episode-level current-weighted coverage $C_{\mathrm{w}}(z)$ is larger for IPPO than for every baseline in all 14 depth–speed settings, by factors of 1.41–1.60 for potential field, 1.39–1.56 for A*, 1.17–1.31 for lawnmower coverage, and 1.95–2.28 for information-greedy control. IPPO has the broadest current-weighted observation but does not have the strongest value in every individual underwater metric.

Appendix Table A.1 resolves these averages by deployment radius and ocean depth. IPPO retains the lowest final error and time-averaged distance at all five surface radii. Underwater, the potential field gives the highest mean experienced current and IPPO gives the largest nominal coverage at every depth.

## 6. Conclusion

This study developed and evaluated a parameter-sharing IPPO controller for coordinated surface–subsurface tropical-cyclone observation with Blue Whale-class SUVs. The operational concept assumes deployment from a mother ship, such as a large research vessel, or from an offshore floating platform such as the Far-Seas Floating Island. Current and future-upgrade speed settings were evaluated separately for the surface and fixed-depth tasks. The event-specific environments combined ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks for 500 western North Pacific TCs. Training used 450 TCs, and evaluation used 50 unseen TCs.

At the surface, globally shared wind observations allowed the fleet to approach the moving TC despite uncertain deployment. The 20 m/s controller produced smaller final and time-averaged tracking distances at every deployment radius, and its final-error advantage grew with deployment error. The 12 m/s controller produced shorter paths and lower speeds. Action filtering further reduced the fleet-mean final tracking error from 313.1 to 266.0 km while shortening paths and reducing peak speed.

At depth, four SUVs sampled one of seven fixed layers using only their local current histories. Relative to random control and passive drift, active IPPO achieved a 2.2–4.6% increase in configuration-mean experienced current, with nominal and current-weighted coverage substantially larger in every depth–speed setting. The traditional-planner comparison showed no single winner across all underwater metrics: potential field obtained the highest mean current, while IPPO obtained the largest nominal and current-weighted coverage. At the surface, IPPO had the lowest final tracking error when averaged over the 40 matched configurations; it was lower than potential field in 35 comparisons and lower than each other planner in all 40.

Under the same 20 h horizon, increasing the submerged speed limit from 1.5 to 2.0 m/s expanded the sampled area at every depth. A faster future SUV offers greater spatial sampling capacity, provided that endurance, control stability, and mission recovery remain feasible.

The present evidence is simulation-based and uses a mission-level horizontal point-mass model. Field validation, operational forecast impact, and calibrated six-degree-of-freedom vehicle dynamics remain future work. The experiments tested generalization to unseen TCs within the simulated western North Pacific environment; they do not establish sim-to-real performance.

## Code and package availability

The code and processed results are available at `https://github.com/DYHAI/SUV-MARL-TC`. The experiments used Python 3.11, PyTorch 2.11.0 (CUDA 12.8 build), Stable-Baselines3 2.9.0, and Gymnasium 1.2.2 for policy training; NumPy 2.4.6 and pandas 2.3.3 for numerical and tabular processing; SciPy 1.17.1, xarray 2026.7.0, netCDF4 1.7.4, and h5py 3.16.0 for analysis and NetCDF processing; and Matplotlib 3.11.1 for visualization. All policy training was performed on CPU.

## Acknowledgements

This project is supported by the National Natural Science Foundation of China (42530402 and 42227901), the National Natural Science Foundation of China Youth Science Fund Project (62603673), the China Postdoctoral Science Foundation (2025M783467), the Project of State Key Laboratory of Satellite Ocean Environment Dynamics (SOEDZZ2527), and the Innovation Group Project of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (311024004).

# A. Reward and IPPO Hyperparameters

Table A.1 reports the traditional-planning comparison with the two surface distance metrics separated by deployment radius and the three subsurface observation metrics separated by depth.

Table A.1: Distance and observation metrics for the traditional planner comparison on the held-out TC segments. Panel A reports the two surface distance metrics by deployment radius, averaged over the four fleet sizes and both speed limits. Panel B reports mean experienced current speed, nominal coverage, and current-weighted coverage at each depth, averaged over the two propulsion limits. Current-weighted coverage is reported in $10^3$ m s$^{-1}$ km$^2$.

**Panel A: surface distance metrics**

| Radius | Metric | IPPO | PF | A* | Lawn | Greedy |
|---|---|---|---|---|---|---|
| 100 km | Final error (km) | **220.8** | 290.4 | 325.8 | 366.1 | 295.2 |
| | Time-averaged distance (km) | **170.6** | 229.9 | 250.5 | 232.9 | 212.0 |
| 200 km | Final error (km) | **236.1** | 295.4 | 330.0 | 394.5 | 310.7 |
| | Time-averaged distance (km) | **197.6** | 248.5 | 267.1 | 285.9 | 244.5 |
| 400 km | Final error (km) | **304.0** | 353.5 | 394.9 | 504.5 | 405.1 |
| | Time-averaged distance (km) | **303.0** | 343.0 | 367.4 | 441.8 | 374.5 |
| 600 km | Final error (km) | **423.9** | 476.1 | 527.6 | 661.2 | 560.1 |
| | Time-averaged distance (km) | **460.8** | 505.1 | 536.7 | 621.7 | 559.1 |
| 800 km | Final error (km) | **596.1** | 651.8 | 707.6 | 847.4 | 759.8 |
| | Time-averaged distance (km) | **650.8** | 694.4 | 725.0 | 809.7 | 756.7 |

**Panel B: subsurface metrics by depth**

| Depth | Metric | IPPO | PF | A* | Lawn | Greedy |
|---|---|---|---|---|---|---|
| 0 m | Mean experienced current (m/s) | 0.534 | **0.577** | 0.548 | 0.536 | 0.551 |
| | Nominal coverage (km$^2$) | **3672** | 2214 | 2445 | 3067 | 1579 |
| | Current-weighted coverage | **2.165** | 1.431 | 1.469 | 1.760 | 0.949 |
| 10 m | Mean experienced current (m/s) | 0.474 | **0.515** | 0.489 | 0.472 | 0.492 |
| | Nominal coverage (km$^2$) | **3729** | 2189 | 2414 | 3020 | 1620 |
| | Current-weighted coverage | **1.926** | 1.252 | 1.290 | 1.518 | 0.872 |
| 20 m | Mean experienced current (m/s) | 0.444 | **0.486** | 0.462 | 0.444 | 0.464 |
| | Nominal coverage (km$^2$) | **3720** | 2152 | 2388 | 2987 | 1632 |
| | Current-weighted coverage | **1.808** | 1.169 | 1.211 | 1.418 | 0.838 |
| 30 m | Mean experienced current (m/s) | 0.409 | **0.452** | 0.429 | 0.410 | 0.431 |
| | Nominal coverage (km$^2$) | **3574** | 2105 | 2329 | 2939 | 1588 |
| | Current-weighted coverage | **1.616** | 1.067 | 1.110 | 1.297 | 0.758 |
| 40 m | Mean experienced current (m/s) | 0.366 | **0.412** | 0.386 | 0.368 | 0.391 |
| | Nominal coverage (km$^2$) | **3330** | 1999 | 2273 | 2876 | 1542 |
| | Current-weighted coverage | **1.373** | 0.924 | 0.977 | 1.144 | 0.669 |
| 50 m | Mean experienced current (m/s) | 0.326 | **0.374** | 0.349 | 0.327 | 0.351 |
| | Nominal coverage (km$^2$) | **3328** | 1876 | 2200 | 2803 | 1470 |
| | Current-weighted coverage | **1.222** | 0.792 | 0.858 | 0.989 | 0.575 |
| 60 m | Mean experienced current (m/s) | 0.302 | **0.350** | 0.325 | 0.301 | 0.324 |
| | Nominal coverage (km$^2$) | **3189** | 1799 | 2120 | 2702 | 1410 |
| | Current-weighted coverage | **1.083** | 0.714 | 0.767 | 0.879 | 0.515 |

Table A.2 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table A.3 lists the shared IPPO network and training hyperparameters.

Table A.2: Surface and subsurface reward, action-filter, and coastal-safety parameters used in the final experiments.

| Parameter | Symbol | Value | Description and applicable task |
|---|---|---|---|
| **Panel A: surface reward and controller parameters** | | | |
| Proximity-cost distance cap | $D_{\text{max}}$ | 300 km | Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments. |
| Distance normalization scale | $D_{\text{scale}}$ | 100 km | Normalization scale used by the proximity and radius-excess reward terms. |
| Near-center threshold | $d_{\text{near}}$ | 100 km | Threshold defining the discrete near-center bonus. |
| Near-center bonus | $b_{\text{near}}$ | 1.0 | Bonus assigned when an SUV is within 100 km of the TC center. |
| Radius-excess threshold | $R_{\text{form}}$ | 100 km | Reference radius used by the additional distance penalty beyond 100 km. |
| Radius-excess weight | $W_{\text{form}}$ | 1.0 | Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center. |
| Action-change weight | $\lambda_u$ | 0.05 | Weight of the squared action change $\|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2$ between successive executed commands. |
| Action-filter coefficient | $\beta$ | 0.8 | Coefficient of $\mathbf{a}_k^{\text{filt}} = (1-\beta)\mathbf{a}_k^{\text{raw}} + \beta\mathbf{a}_{k-1}^{\text{filt}}$. This is a controller setting and is not included in the reward. |
| Surface reward clipping range | $[r_{\text{min}}, r_{\text{max}}]$ | $[-5, 2]$ | Clipping range applied to the team-averaged surface reward. |
| **Panel B: underwater active-control comparison parameters** | | | |
| Current-speed normalization | $V_{\text{norm}}$ | 1.0 m/s | Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\text{max}})$. |
| Current-reward cap | $R_{\text{max}}$ | 2.0 | Upper bound of the normalized current-speed reward. |
| Downstream-alignment weight | $\beta_{\parallel}$ | 0.5 | Weight of the positive propulsion–current alignment term. |
| Underwater reward clipping range | $[r_{\text{min}}, r_{\text{max}}]$ | $[-5, 5]$ | Clipping range applied to the team-averaged underwater reward during active control training. |
| **Panel C: coastal-safety parameters by stage** | | | |
| Land penalty | $P_{\text{land}}$ | 20 | Penalty applied when an SUV enters a land cell in either stage. |
| Nearshore buffer distance | $D_{\text{buf}}$ | 100 km | Width of the soft nearshore penalty buffer in both stages. |
| Nearshore penalty weight | $W_b$ | 8.0 | Maximum soft-penalty weight within the 100 km nearshore buffer in either stage. |
| Minimum deployment offshore distance | $D_{\text{deploy}}$ | 100 km | Minimum coastline distance required by offshore-only rejection sampling during surface deployment; it is not applied to subsurface initialization. |
| Action-filter safety margin | $D_{\text{filter}}$ | 100 km | Surface-stage distance below which the onshore component of the commanded action is removed. The subsurface comparison does not use this filter. |

Note: the final subsurface comparison consists of active IPPO, random control, and passive drift. Passive drift does not optimize a learned propulsion policy or reward.

Table A.3: Parameter-sharing IPPO network, training, and deployment hyperparameters used in the final surface and subsurface experiments.

| Parameter | Symbol | Value | Description |
|---|---|---|---|
| Algorithm | – | PPO | Stable-Baselines3 PPO with parameter sharing and independent per-agent action sampling. |
| Return aggregation | – | Team average | The environment reward is averaged across the $N$ SUVs before computing the shared PPO return. |
| Optimizer | – | Adam | Adam optimizer used for the actor and critic parameters. |
| Policy network | – | $2 \times 256$ MLP | Actor: Linear($d_{\text{obs}}, 256$)–Tanh– Linear($256, 256$)–Tanh–Linear($256, d_{\text{act}}$). |
| Value network | – | $2 \times 256$ MLP | Critic: Linear($d_{\text{obs}}, 256$)–Tanh– Linear($256, 256$)–Tanh–Linear($256, 1$). Per-agent values are summed during training. |
| Per-agent observation dimension | $d_{\text{obs}}$ | 16/28/40/52; 19 | Surface observations for 4/8/12/16 SUVs are 16/28/40/52, respectively. The four-SUV underwater active-control comparison uses $d_{\text{obs}} = 19$. |
| Surface observation | – | Wind at 10 m above sea level | Using ideal surface communication, each SUV receives the current wind speed and components at 10 m above the sea surface observed by every SUV in the fleet, together with its normalized ground velocity and $[\cos\psi, \sin\psi]$. |
| Underwater observation | – | Current only | With no inter-vehicle communication, the active IPPO controller observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history, its own normalized propulsion velocity, and $[\cos\psi, \sin\psi]$. |
| Action dimension | $d_{\text{act}}$ | 2 per SUV | Continuous east–north velocity command in $[-1, 1]^2$, followed by circular projection to respect the maximum speed. |
| Parallel environments | $N_{\text{env}}$ | 4; 4 or 5 | All surface runs and three subsurface runs used four vectorized training environments; the remaining 137 subsurface runs used five. |
| Rollout length | $n_{\text{steps}}$ | 2048 | Number of steps collected per environment before each PPO update. |
| Mini-batch size | $B$ | 256 | Mini-batch size used during each gradient update. |
| Optimization epochs | $N_{\text{epochs}}$ | 10 | Number of passes over each rollout. |
| Learning rate | $\alpha$ | $3 \times 10^{-4}$ | Adam learning rate used for the actor and critic. |
| Discount factor | $\gamma$ | 0.99 | Reward discount factor. |
| GAE parameter | $\lambda_{\text{GAE}}$ | 0.95 | Generalized advantage estimation applied to the team-average reward. |
| Policy clipping range | $\epsilon$ | 0.2 | Clipping range of the PPO probability ratio. |
| Value clipping range | $\epsilon_v$ | 0.2 | Clipping range of the value-function update. |
| Entropy coefficient | $c_{\text{ent}}$ | 0.01 | Entropy regularization coefficient. |
| Value-loss coefficient | $c_v$ | 0.5 | Weight of the value-function loss. |
| Maximum gradient norm | $g_{\max}$ | 0.5 | Global gradient clipping threshold. |
| Training steps | $T$ | $2 \times 10^6$ | Nominal environment steps per training run. |
| Independent seeds | – | 0–9 | Ten random seeds for each selected surface or active-control configuration. |
| Surface training forecast-error radius | $R_{\text{train}}$ | 0–800 km | The candidate surface formation-center radial offset is sampled uniformly in $[0, 800]$ km, with a uniformly sampled bearing. This models the assumed upper bound of TC track-forecast error; offshore-only rejection sampling is then applied as part of the safety layer. |
| Surface validation center offsets | $R_{\text{test}}$ | 100/200/400/600/800 km | Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement. |
| Underwater initialization radius | $R_{0,\text{u}}$ | ≤200 km | At the end time of each surface segment, the separate subsurface episode initializes four vehicles with radial offsets sampled uniformly in $[0, 200]$ km and uniformly sampled bearings. Surface endpoint positions are not transferred. The 0–800 km forecast-error protocol is surface-only. |
| Decision interval | $\Delta t$ | 10 min | Environmental observation and action interval. |
| Surface formation radius | $R_{\text{ring}}$ | 100 km | Surface SUVs use equal angular spacing of $360°/N$ on a 100 km ring. |
| Surface fleet sizes | $N$ | 4/8/12/16 | Number of SUVs in the final surface configurations. |
| Underwater fleet size | $N$ | 4 | Number of SUVs in each active-IPPO, random-control, and passive-drift case. |
| Underwater depths below the sea surface | $z$ | 0/10/20/30/40/50/60 m | Seven ocean-current observation depths used in the final subsurface matrix. |
| Underwater methods | – | Active IPPO; random; drift | Three paired methods evaluated on the same episodes and deployments. |
| Underwater duration | $H$ | 20 h | Fixed simulation horizon for the final subsurface comparison; this is not an endurance specification. |
| Surface ground-speed limits | $V^{\text{s}}_{\max}$ | 12 or 20 m/s | Two commanded ground-speed envelopes; these are mission-level scenarios, not independently measured certifications. |
| Submerged speed limits | $V^{\text{u}}_{\max}$ | 1.5 or 2.0 m/s | Two propulsion-speed limits after unit-disk action projection used for the active-control comparison. |
| Compute device | – | CPU | All reported IPPO experiments were trained on CPU. |

## B. Supplementary Surface Formation Trajectory

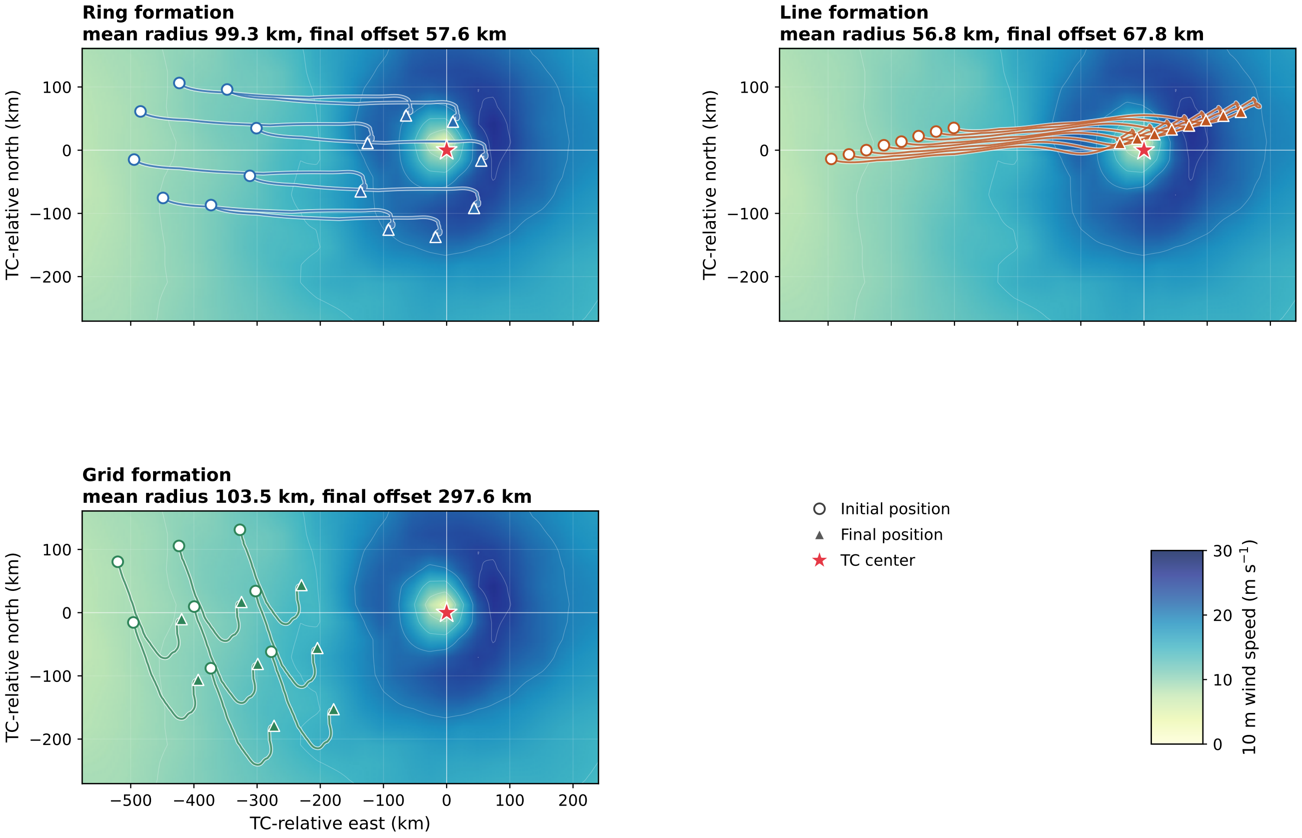


Figure B.1: Representative surface trajectory comparison for ring, line, and grid formations with eight SUVs at 12 m/s. Open circles denote initial positions, triangles denote final positions, and the red star denotes the TC center. The color background and contours show the spatially interpolated ERA5 10 m wind-speed field at the final simulation step. Axes are relative to the instantaneous TC center.

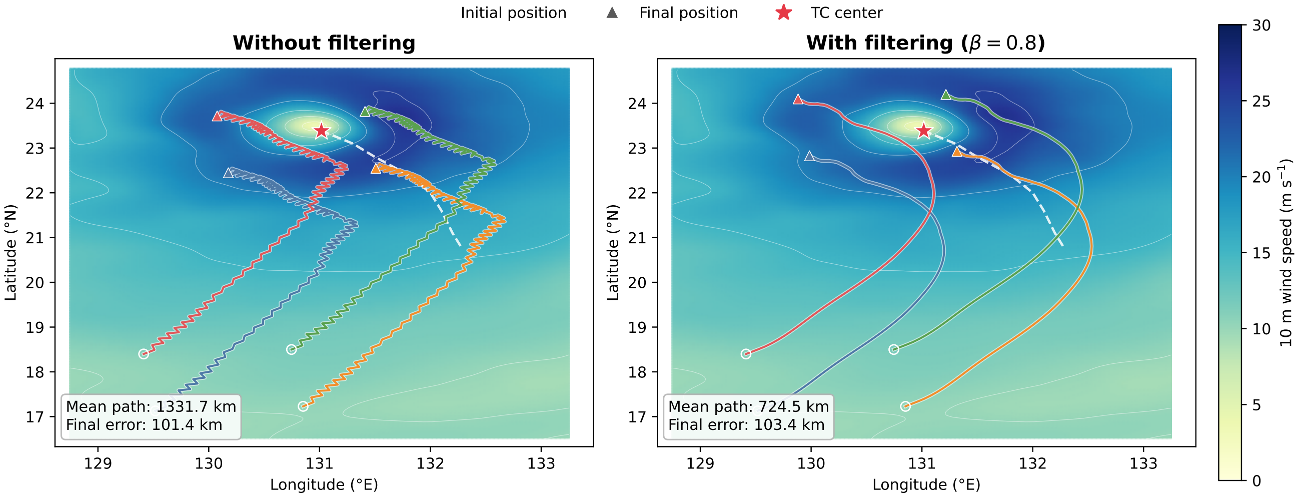


Figure B.2: Representative effect of execution-layer filtering on surface paths for four SUVs under the same trained policy, initial positions, deployment radius, and KHANUN (2023) segment. The left panel uses the raw policy command, whereas the right panel applies the low-pass filter with $\beta = 0.8$. Open circles and triangles mark initial and final positions, respectively, and the red star marks the TC center.

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